



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)

Unit 1 – Fundamentals of OOP

Topic – 1.5 Data Types in C++

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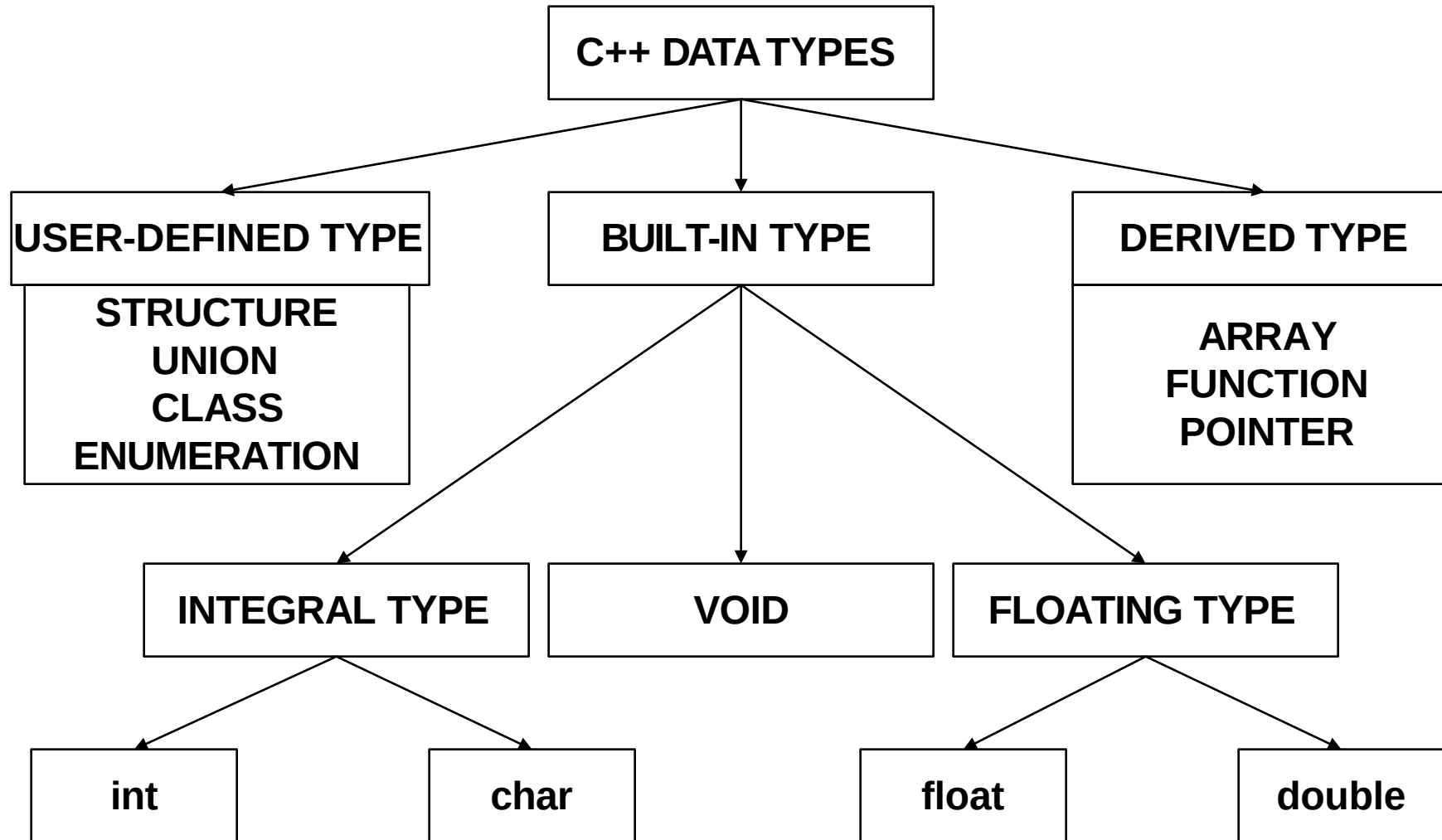
Data type

- A data type specifies the type of value a variable can hold.
- General syntax of declaration of data type is:

`data_type variable_name;`

- Example:

`int a;`

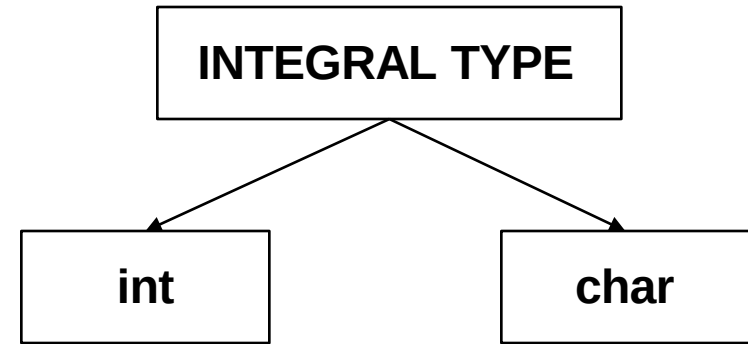




INTEGRAL DATATYPE

INTEGER DATATYPE:

- Keyword: int
- Storage space: two bytes
- Range of int: -32768 to 32767



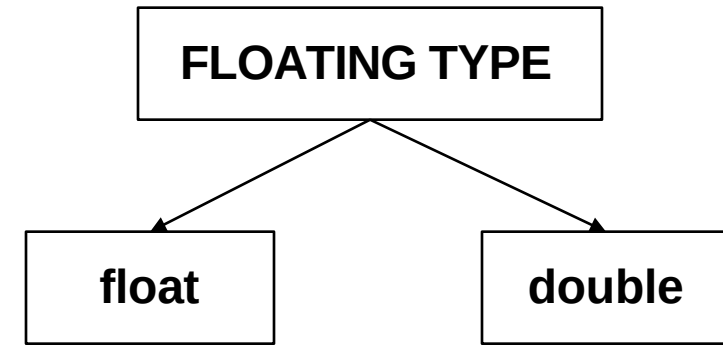
CHARACTER DATATYPE:

- Keyword: char
- Storage space: 1 byte
- Range of char: -128 to 127



FLOAT DATATYPE

- Keyword: float
- Storage space: 4 bytes
- Range of float: $3.4e-38$ to $3.4e+38$.





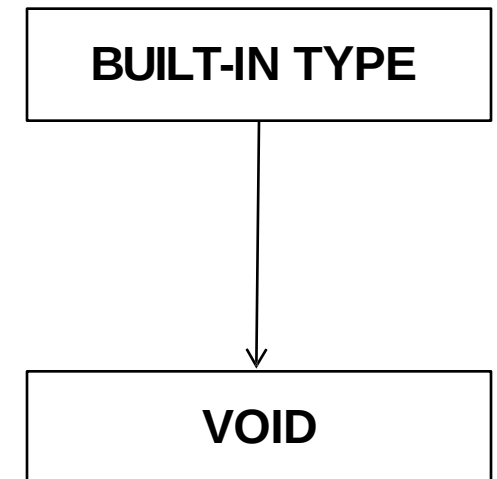
VOID DATATYPE

- Type void was introduced in ANSI C.

Two normal use of void:

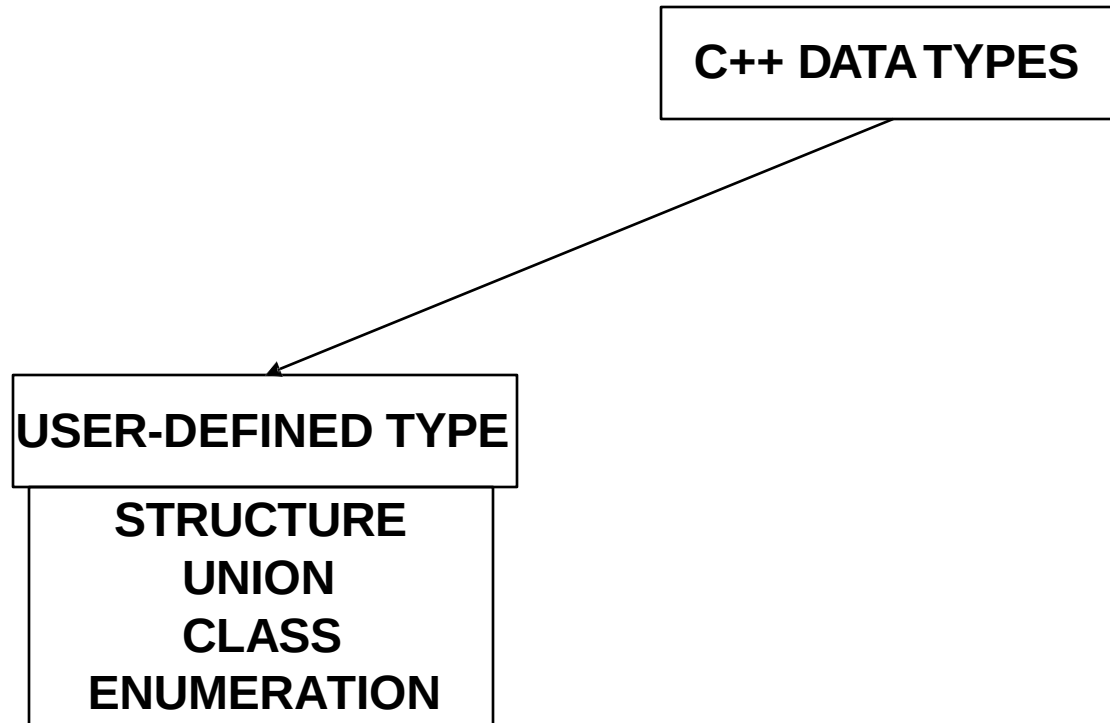
- To specify the return type of a function when it is not returning any value.
- To indicate an empty argument list to a function.

Eg:- void function-name (void)





User Defined Type





User Defined Type

- STRUCTURES:

- Structures are used for grouping together elements with dissimilar types.
- The general syntax of a structure:-

```
struct struct_name  
{  
    Data_type member1;  
    . . . . .  
};
```




User Defined Type

- Unions:
 - Unions are conceptually similar to structures as they allow us to group together dissimilar type elements inside a single unit.

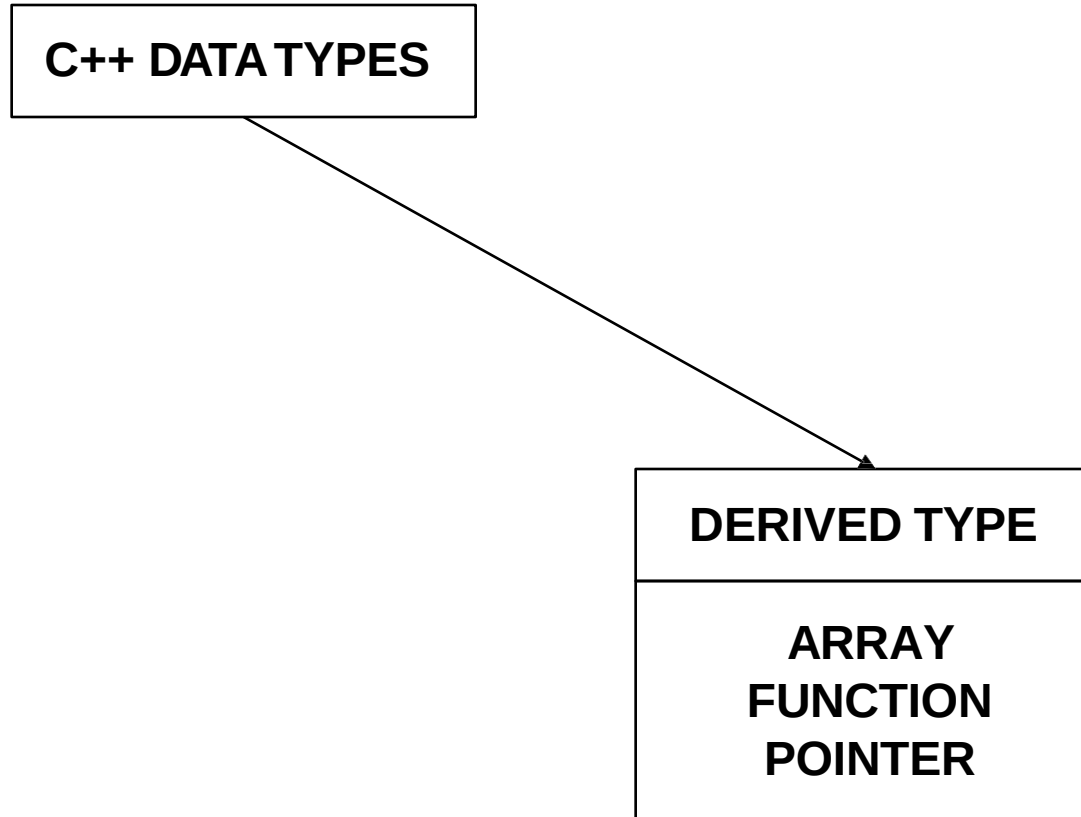


User Defined Type

- CLASSES:
 - The entire set of data and functions can be made a user-defined data type with the help of a class.
 - Once a class has been defined, we can create any number of objects belonging to that class.
 - Example: fruit mango;
 - Created an object mango belonging to the class fruit.



Derived Type





Derived Type

- ARRAYS:

- An array is a fixed sequence collection of elements of same data type that share a common name.

- General syntax is:

`data_type array_name[size];`

- Example:

`int a[10];`



Derived Type

- FUNCTIONS:

- A function is a group of statements that together perform a task.
- Every c++ program has at least one function, which is main().

```
void main()  
{  
    Statements;  
}
```



Derived Type

- POINTERS:

- A pointer is a variable that holds a memory address.
- General syntax to declare pointer is:-

Data_type *pointer_name;

- Example:

int *p;

Thank You..